

Physics-Guided Flow-Map Matching for Precipitation Nowcasting

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Abstract. Precipitation nowcasting predicts short-term rainfall from radar observations and is critical for flood and disaster early warning. Deterministic models minimize a pixel loss and are driven toward the conditional mean, which blurs the high-intensity cells that matter most, while generative models that add a stochastic residual on top of a deterministic backbone inherit the same blur. We propose Physics-Guided Flow-Map Matching (PG-FMM), a generative nowcaster that decouples predictable advection from uncertain detail. A frozen Lagrangian advection prior transports the radar field and supplies an interpretable motion forecast, and a flow-map generative head—conditioned on the past frames and the prior rollout rather than summed onto it—renders calibrated stochastic detail with few-step sampling. Because the physical forecast is used only as guidance, the head replaces blurred structure with sharp detail instead of inheriting it. Under a unified evaluation protocol over four radar benchmarks (SEVIR, MeteoNet, Shanghai, and CIKM), PG-FMM outperforms the strongest published baseline on 18 of 24 metrics, with the largest gains at the heavy-rain thresholds.

Keywords: Precipitation nowcasting · Generative models · Flow matching · Spatiotemporal forecasting

1 Introduction

Precipitation nowcasting predicts rainfall at high spatial and temporal resolution over the coming hours from a short history of radar observations, and it underpins flood early warning, disaster response, water management, and the safe operation of transport and energy infrastructure [2, 39, 56]. Its importance is growing: a warming climate makes intense convective rainfall more frequent and more damaging, and the short lead time of a nowcast is often what separates an effective response from a costly one. Given a sequence of past radar frames, the task is to predict a sequence of future reflectivity fields. As heavy-rainfall events become more common, the value of a nowcast increasingly rests on the high-intensity regime, where the goal is to forecast where and how strongly convective cells will develop rather than merely whether it will rain.

This regime is also the hardest, because small-scale convective growth and decay are only partly predictable. Deterministic models minimize a per-pixel regression loss and are therefore driven toward the conditional mean of the future,

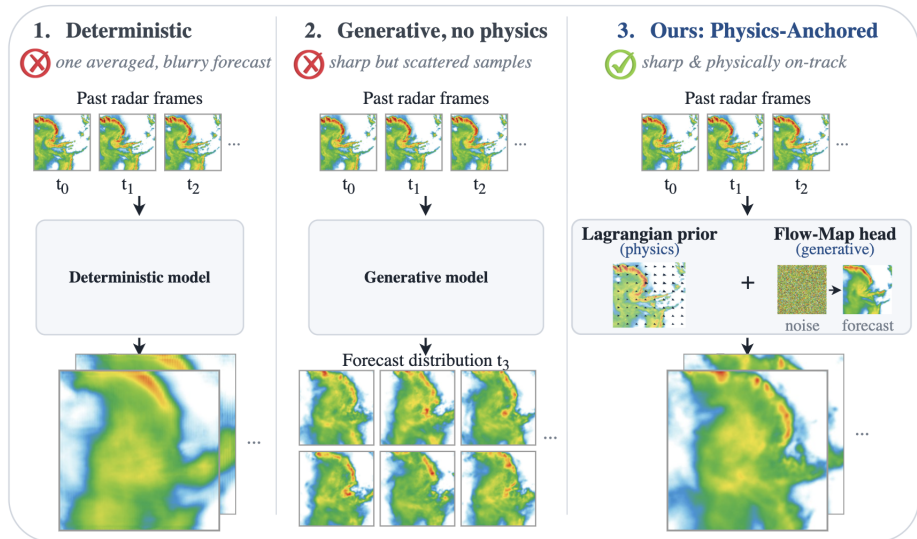


Fig. 1: Three families of precipitation nowcasters. (1) Deterministic models minimize a pixel loss and are driven toward the conditional mean, producing a single averaged forecast that smooths away high-intensity cells. (2) Purely generative models sample sharp fields but, without a physical prior, scatter convection off the observed storm track. (3) Our physics-guided design conditions a generative flow-map head on a frozen Lagrangian advection prior, keeping forecasts sharp and on-track.

so their forecasts blur [11, 13, 42, 43, 50]. The blur erases the sharp, high-intensity cores that matter most, and detection skill at the heaviest thresholds decays toward zero as the lead time grows. This failure mode is shared even by strong recent predictors that add physical structure or operate in the frequency domain, such as PhyDNet [22] and AlphaPre [24]: a mean-seeking objective smooths extremes regardless of the backbone. A faithful nowcaster therefore has to be generative, sampling sharp and plausible fields rather than averaging them.

Existing generative nowcasters address sharpness but leave two gaps. Many recent diffusion-based methods follow a residual-anchor recipe, adding a generated stochastic residual on top of a deterministic backbone [12, 14, 55]; because that backbone is trained with a pixel loss, it carries conditional-mean blur, and the sum inherits the blur at long lead times. Purely generative models, including GAN- and flow-based nowcasters [35, 39], avoid the anchor but discard the predictable part of the dynamics—the advective motion that a simple physical model already captures well—and must relearn it from data, which leaves convection prone to drifting off the observed storm track. What is missing is a way to retain the motion skill of a physical model while letting a generative model render the unpredictable detail, without either contaminating the other. Figure 1 contrasts these three families.

In this study we propose Physics-Guided Flow-Map Matching (PG-FMM), a generative nowcaster that guides a flow-map generative head with a frozen physical motion model. A lightweight Lagrangian advection prior is pre-trained once and then frozen; it rolls the observed field forward through a learned velocity and source term and supplies an interpretable motion forecast. A flow-map generative head then produces the future sequence conditioned on both the history and this prior rollout. The decisive design choice—and the reason this particular formulation is needed—is that the prior enters only as a conditioning signal and is never added to the output: the head is free to replace blurred structure with sharp detail instead of inheriting it, which no residual-anchor model can do, while the physical rollout still anchors the convection to a plausible trajectory, which no purely generative model guarantees. Casting the generator as a two-time flow map, rather than an instantaneous velocity field or a long diffusion chain, lets it generate in a handful of network evaluations and keeps the formulation faithful to the few-step transport view of recent consistency models [6, 44]. This separation of predictable motion from stochastic detail follows the decoupling logic we developed for spatiotemporal precipitation modeling in prior work [30].

Across four standard radar benchmarks and six metrics each, PG-FMM outperforms the strongest published baseline, AlphaPre [24], on 18 of the 24 metrics, and the margins are largest precisely at the heavy-rain thresholds where mean-seeking methods fail. The main contributions of this work are summarized as follows.

- We propose PG-FMM, which guides a generative flow-map head with a frozen Lagrangian advection prior used as conditioning rather than as an additive residual, so that conditional-mean blur is never transmitted to the prediction.
- We introduce, to our knowledge, the first flow-map-matching formulation for precipitation nowcasting, coupling a learned semi-Lagrangian advection prior with a two-time transport head trained under a composition-consistency objective.
- We show on four radar benchmarks (SEVIR, MeteoNet, Shanghai, and CIKM) that PG-FMM surpasses AlphaPre on 18 of 24 metrics, with the largest gains at the heavy-rain thresholds, and we isolate the source of the gains with component ablations.

2 Related Work

2.1 Precipitation nowcasting

Deep learning for radar nowcasting has been surveyed extensively from both the time-series and the weather-modeling perspectives [2, 41], and it inherits much of its machinery from video prediction [29, 32]. Early deterministic models cast nowcasting as next-frame regression with recurrent or convolutional sequence networks: ConvLSTM [42] and its trajectory variant TrajGRU [43], the PredRNN family [49–51], MotionRNN [52], and the efficient convolutional predictor SimVP [11]. Transformer and operator backbones such as Earthformer [13]

and Rainformer [4], and physically structured predictors such as PhyDNet [22] and MAU [8], improve average skill, and the recent amplitude–phase model AlphaPre [24] is the strongest deterministic predictor on the benchmarks we use. Because a squared- or absolute-error objective is minimized by the conditional mean, all of these models tend to blur, and the blur is most damaging at high reflectivity where sharp convective cells are smoothed away.

Generative nowcasters instead learn a conditional distribution over futures and sample from it. DGMR [39] uses a conditional GAN, and NowcastNet [56] couples a deterministic evolution operator with a generative network. Diffusion and latent-diffusion models have since become dominant: PreDiff [12] and LD-Cast [23] perform latent diffusion, CasCast [14] cascades a deterministic prediction with a latent diffusion refiner, and DiffCast [55] adds a stochastic residual on top of an arbitrary deterministic backbone. We refer to the latter recipe as a residual anchor: the forecast is a deterministic prediction plus a generated residual. Because the deterministic part is trained with a pixel loss, it carries conditional-mean blur that the sum cannot remove, particularly at long lead times. Related video-diffusion predictors such as MCVD [48] and STRPM [7], and diffusion models for the broader weather-forecasting problem [31, 36, 46], share the same generative motivation at different spatial and temporal scales.

2.2 Flow matching and flow maps

Continuous-time generative modeling has matured rapidly, and recent surveys [26, 54] cover the progression from score-based diffusion [15, 17, 45] to simulation-free transport. Flow Matching [25], Rectified Flow [28], and Stochastic Interpolants [1] regress an instantaneous velocity field whose probability-flow ODE must be integrated with many steps at sampling time. Flow-Map Matching [6] instead learns the finite-time flow map—the two-time transport operator—so that a sample is produced in a few, or even one, network evaluations; Consistency Models [44], Consistency Trajectory Models [18], and Shortcut Models [10] reach the same few-step regime through self-consistency, trajectory matching, and step-size conditioning, and distillation offers a complementary route [40]. These transport formulations also include data-to-data diffusion bridges such as I²SB [27] and DDBM [57]. In nowcasting, FlowCast [35] applies conditional flow matching in a latent space. We adopt a flow-map head because its few-step sampling makes a multi-member inference ensemble affordable, which is otherwise prohibitive for long diffusion chains.

2.3 Physics-informed and Lagrangian forecasting

Physics-informed machine learning injects physical knowledge into data-driven models through soft PDE-residual penalties, structured operators, or explicit dynamics, and is reviewed comprehensively by Karniadakis *et al.* [16, 38]. In precipitation nowcasting, a long operational tradition extrapolates the radar field along estimated motion, as in the optical-flow libraries PySTEPS [37] and rainymotion [3]; such Lagrangian advection is accurate for short horizons but

cannot represent growth or decay. Learning-based work reintroduces the missing source term while keeping the motion explicit: NowcastNet [56] learns a neural evolution operator with an additive intensity update, PhyDNet [22] disentangles physical dynamics from residual factors in a latent space, and the differentiable Lagrangian network LUPIN [34] separates advection from an advection-free growth/decay branch. Beyond nowcasting, large data-driven weather models such as FourCastNet [33], Pangu-Weather [5], GraphCast [20], and Neural-GCM [19] show that physically informed architectures scale to global forecasting. Our prior follows the explicit warp-and-source route but keeps the physics module lightweight, interpretable, and, after a short pre-training stage, frozen, so that it supplies motion skill as a conditioning signal rather than as a constraint on the generative head.

3 Preliminaries

3.1 Problem setting

We address radar precipitation nowcasting. A model observes a history of T_{in} past radar frames and predicts the next T_{out} frames, where each frame is a single-channel reflectivity field $R \in \mathbb{R}^{H \times W}$ on a spatial grid of size $H \times W$ normalized to $[0, 1]$. We write the observed history as $R_{1:T_{\text{in}}}$ and the prediction target as

$$x_0 = R_{T_{\text{in}}+1:T_{\text{in}}+T_{\text{out}}} \in \mathbb{R}^{T_{\text{out}} \times H \times W}, \quad (1)$$

where the T_{out} future frames are stacked along the channel axis for the convolutional networks. The horizon lengths $T_{\text{in}}, T_{\text{out}}$, the grid size $H \times W$, and the per-dataset intensity scale used to recover physical units are dataset-specific and are listed in Sec. 5.

Because convective growth and decay are only partly determined by the recent past, the future is genuinely uncertain. We therefore model the conditional distribution $p(x_0 | R_{1:T_{\text{in}}})$ and draw an ensemble of forecasts from it rather than predicting a single field. A good forecast combines high detection skill at the operational intensity thresholds—especially the heavy-rain thresholds—with low pixel error and sharp, non-blurred structure. We report the Critical Success Index (CSI) per threshold and averaged, the Heidke Skill Score (HSS), SSIM, and MSE, with formal definitions deferred to Sec. 5; the primary metrics are the mean CSI and the high-threshold CSI. We consider two-dimensional single-channel radar and do not address three-dimensional or multi-channel observations.

3.2 Flow matching and flow maps

Continuous-time generative models transport a simple base density to the data distribution along an interpolant between a noise sample $x_1 \sim \mathcal{N}(0, I)$ and a data sample x_0 ,

$$x_\tau = (1 - \tau) x_0 + \tau x_1, \quad (2)$$

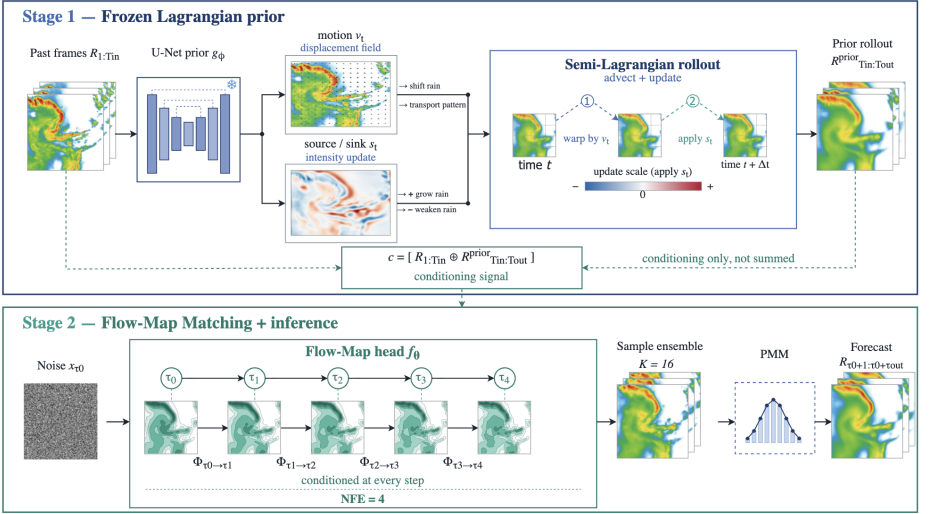


Fig. 2: Overview of the proposed method. A frozen Lagrangian advection prior g_ϕ turns the observed frames into an interpretable motion forecast through a learned velocity and source rollout $R^{(t)} = \text{clip}(\text{warp}(R^{(t-1)}, v_t) + s_t)$. A Flow-Map Matching head f_θ , conditioned on the history and the prior rollout and never summed onto the output, maps noise to the future sequence with a two-time transport operator $\Phi_{\tau_i \rightarrow \tau_{i+1}}$ in four steps. At inference a 16-member probability-matched-mean ensemble restores per-pixel intensity. Stage 1 (top) is frozen; Stage 2 (bottom) is trained, and the prior enters only as a conditioning signal, never summed onto the output.

where $\tau \in [0, 1]$ indexes the path from data ($\tau=0$) to noise ($\tau=1$). Flow matching learns an instantaneous velocity field $v_\theta(x_\tau, \tau)$ whose probability-flow ODE carries the base density to the data density, and sampling integrates this ODE, which typically requires many function evaluations.

Flow-Map Matching instead learns the two-time solution operator of the same ODE. The flow map $\Phi_{t \rightarrow r}$ takes a state at time t directly to the corresponding state at an earlier time r ,

$$\Phi_{s \rightarrow r} \circ \Phi_{t \rightarrow s} = \Phi_{t \rightarrow r}, \quad (3)$$

where $0 \leq r < s < t \leq 1$ and the semigroup identity holds because a flow map is a solution operator. Enforcing this composition consistency during training regularizes the learned operator, and because the map spans a finite time interval a sample is produced in a few jumps rather than many small steps. Our generative head (Sec. 4) instantiates a conditional flow map of this form.

4 Method

4.1 Overview

PG-FMM forecasts precipitation by separating the predictable advective motion of the radar field from the uncertain small-scale detail, so that neither part contaminates the other. The model has two modules (Fig. 2): a frozen Lagrangian advection prior g_ϕ (Sec. 4.2) that turns the observed history into an interpretable motion forecast, and a flow-map generative head f_θ (Sec. 4.3) that, conditioned on the history and the prior rollout, samples the future sequence. At inference we draw a small ensemble and aggregate it with a probability-matched mean (Sec. 4.4).

The prior is used only as conditioning and is never added to the output. This is the property that residual-anchor nowcasters such as DiffCast [55] and CasCast [14] lack: they write the forecast as a deterministic backbone plus a generated residual, so the conditional-mean blur of the backbone is carried into the sum at long lead times. By passing the physical forecast as a conditioning signal instead, we let the generative head replace the prior’s blurred structure with sharp detail while still exploiting its motion skill. The decoupling is agnostic to the particular modules: any per-step motion model and any conditional generative head could be substituted; our instantiation pairs a learned warp-and-source prior with a flow-map head whose few-step sampling makes the inference ensemble affordable.

4.2 Lagrangian advection prior

Precipitation evolves partly through predictable transport and partly through stochastic growth and decay, and a cheap physical model can already account for the first part. The prior captures this predictable motion explicitly, so the generative head only has to model the residual uncertainty rather than relearn bulk advection from scratch. A U-Net g_ϕ ingests the stacked history $R_{1:T_{\text{in}}}$ and emits, for each future step t , a dense velocity field $v_t \in \mathbb{R}^{2 \times H \times W}$ and a scalar source/sink field $s_t \in \mathbb{R}^{H \times W}$. Both outputs are bounded for stability,

$$v_t = d_{\max} \tanh(\tilde{v}_t), \quad s_t = c_s \tanh(\tilde{s}_t), \quad (4)$$

where $\tilde{v}_t, \tilde{s}_t$ are the raw network outputs, $d_{\max}$ caps the per-step displacement, and c_s caps the intensity update. Starting from the last observed frame $R^{(0)} = R_{T_{\text{in}}}$, the forecast is rolled out autoregressively,

$$R^{(t)} = \text{clip}_{[0,1]} \left(\text{warp} \left(R^{(t-1)}, v_t \right) + s_t \right), \quad t = 1, \dots, T_{\text{out}}, \quad (5)$$

where $\text{warp}(\cdot, v)$ is backward bilinear sampling—each output pixel reads from the location displaced by v on a normalized grid with border padding—and the additive source s_t lets cells intensify or dissipate, which pure advection cannot represent.

The prior is trained alone, before the generative head, to forecast the future frames under a composite objective that balances intensity-weighted reconstruction, heavy-rain detection, and physical regularity,

$$\mathcal{L}_{\text{prior}} = \ell_1^w + \lambda_{\text{mse}} \text{MSE} + \lambda_{\text{csi}} \mathcal{L}_{\text{soft-CSI}} + \lambda_v \|\nabla v\|_1 + \lambda_s \|s\|_1, \quad (6)$$

where ℓ_1^w is an ℓ_1 term with pixels weighted toward rain, $\mathcal{L}_{\text{soft-CSI}}$ is a differentiable detection reward, $\|\nabla v\|_1$ is a first-order smoothness penalty on the motion field, and $\|s\|_1$ is a sparsity penalty that encourages motion to explain as much of the change as possible; the weights $\lambda_\bullet$ are given in Sec. 5. After this stage the prior is frozen and receives no gradient during generative training or at inference.

4.3 Flow-Map Matching generative head

A deterministic prior cannot supply the sharp, calibrated detail that distinguishes a plausible heavy-rain forecast from its blurred mean, so the second module is generative. It models the conditional distribution of the future frames given the history and the prior rollout, and it is realized as a flow map—rather than an instantaneous velocity field—so that sampling needs only a few network evaluations, which in turn makes the inference ensemble (Sec. 4.4) affordable. We treat the future sequence x_0 as the data endpoint of a linear stochastic interpolant [1] between Gaussian noise $x_1 \sim \mathcal{N}(0, I)$ and data, as in Eq. (2). The conditioning c concatenates, along the channel axis, the observed history $R_{1:T_{\text{in}}}$ and the frozen prior rollout $R^{(1:T_{\text{out}})}$ from Eq. (5); it is appended to the network input at every evaluation and, again, never added to the output.

Instead of regressing the instantaneous velocity of the interpolant, the network learns its flow map: $f_\theta(x_t, c; t, r)$ predicts the state x_r directly from the state x_t for $0 \leq r < t \leq 1$ [6]. The two times enter both through a timestep embedding and as two constant channel planes (t, r) , making the operator explicitly two-time. The backbone is a U-Net with self-attention at its coarsest scale; its width and depth are reported in Sec. 5. The primary loss regresses the operator to the true interpolant endpoint,

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{x_0, x_1, (t, r)} \|f_\theta(x_t, c; t, r) - x_r\|_2^2, \quad (7)$$

where the times satisfy a minimum gap $t - r \geq \delta_{\text{min}}$ and, with probability p_0 , we force $r=0$ so the network also learns the direct noise-to-data map used at the first sampling step. To make Φ a genuine transport operator we add a composition-consistency regularizer that enforces the semigroup identity of Eq. (3): for a random intermediate time $s \in (r, t)$, composing $t \rightarrow s$ then $s \rightarrow r$ must reproduce the direct map $t \rightarrow r$,

$$\begin{aligned} \mathcal{L}_{\text{CC}} = & \underbrace{\frac{1}{2} \|\Phi_{t \rightarrow s}(x_t) - x_s\|_2^2 + \frac{1}{2} \|\Phi_{s \rightarrow r}(\Phi_{t \rightarrow s}(x_t)) - x_r\|_2^2}_{\text{sub-map fidelity}} \\ & + \underbrace{\|\Phi_{s \rightarrow r}(\Phi_{t \rightarrow s}(x_t)) - \Phi_{t \rightarrow r}(x_t)\|_2^2}_{\text{self-consistency}}, \end{aligned} \quad (8)$$

where the intermediate prediction is detached before the second map. The total objective is $\mathcal{L}_{\text{FM}} + \lambda_{\text{CC}} \mathcal{L}_{\text{CC}}$, with the schedule for λ_{CC} given in Sec. 5; in our setting the consistency term acts as a stabilization and fidelity regularizer.

4.4 Inference: few-step sampling and ensemble

Because f_θ is a transport operator, a forecast is produced with a small number of function evaluations (NFE). On a decreasing time grid $1=\tau_0 > \tau_1 > \dots > \tau_N=0$, we initialize $x_{\tau_0} \sim \mathcal{N}(0, I)$ and iterate

$$x_{\tau_{i+1}} = f_\theta(x_{\tau_i}, c; \tau_i, \tau_{i+1}), \quad i = 0, \dots, N-1, \quad (9)$$

returning $\text{clip}_{[0,1]}(x_{\tau_N})$, where N is the number of sampling steps and we use exponential-moving-average weights of the trained network. A single draw is sharp but noisy. Few-step sampling makes it cheap to draw an ensemble of K samples and combine them with a Probability-Matched Mean (PMM) [9]: the ensemble-mean field fixes the spatial pattern, and its sorted values are replaced by the rank-matched values of the pooled member distribution to restore intensity. The result keeps the ensemble’s well-placed structure while recovering the high-amplitude peaks that a plain mean smooths away, which is decisive for high-threshold CSI. We report the full PMM for verification metrics and, for qualitative figures, a single sample, which is visually sharper; the operating-point values N and K are given in Sec. 5.

5 Experiments

5.1 Datasets and protocol

We evaluate on four standard radar precipitation benchmarks that span different regions and sensors: SEVIR (United States) [47], MeteoNet (France) [21], Shanghai Radar (China), and CIKM. Following the AlphaPre protocol [24], every frame is resized to $H \times W = 128 \times 128$ and normalized to $[0, 1]$, and we predict T_{out} future frames from $T_{\text{in}}=5$ past frames, with $T_{\text{out}}=20$ on SEVIR, MeteoNet, and Shanghai and $T_{\text{out}}=10$ on CIKM. SEVIR provides 12,182 training and 8,096 test sequences; MeteoNet 6,308 / 1,310; Shanghai 1,381 / 526 (a 10% tail of the official 1,534 training sequences is held out for validation, with no test leakage); and CIKM 1,000 / 4,000. We use each dataset’s native intensity scale and evaluation thresholds and run the same evaluator as the baselines, so all numbers are directly comparable.

5.2 Implementation details

The two modules are trained in sequence with AdamW (learning rate 10^{-4} , bf16, gradient clipping 0.5) and exponential-moving-average weights with decay $\rho=0.999$. In the Lagrangian prior (Eqs. (4) and (6)) the maximum displacement is $d_{\text{max}}=8$ pixels and the source scale is $c_s=0.25$, and the loss weights are

$\lambda_{\text{mse}}=0.25$, $\lambda_{\text{csi}}=0.02$, $\lambda_v=0.01$, and $\lambda_s=10^{-3}$. The flow-map head is a U-Net of base width 128 with channel multipliers [1, 2, 3, 4] and two residual blocks per level, reaching a coarsest resolution of 16×16 where self-attention is applied ($\sim 85\text{M}$ trainable parameters; $\sim 101\text{M}$ including the frozen prior at inference). For the flow-map objective (Eq. (7)) the minimum time gap is $\delta_{\text{min}}=0.05$ and the direct-map probability is $p_0=0.5$; the composition-consistency weight λ_{CC} is ramped linearly from 0 to 0.04 between steps 12k and 30k. The prior is trained for 50k steps on SEVIR / 30k on MeteoNet and then frozen; the head is trained for up to 297.6k steps on SEVIR / 154k on MeteoNet at batch size 8 with a 5k-step warm-up. At inference we use $N=4$ sampling steps and a $K=16$ -member probability-matched-mean ensemble. Training and evaluation run on a single AWS g7e.4xlarge instance with one NVIDIA GB202 GPU (96 GB VRAM); one MeteoNet run (154k steps) takes ~ 11.5 h, and SEVIR test evaluation runs at ~ 7 min per 1,310 sequences at the 16-member operating point. We select checkpoints by validation skill.

5.3 Baselines and metrics

We compare against the baselines reported in the AlphaPre study under the same protocol, including deterministic predictors (ConvGRU [43], MAU [8], SimVP [11], FourCastNet [33], Earthformer [13], PhyDNet [22], EarthFarseer [53]) and the strongest recent deterministic and generative methods, NowcastNet [56], DiffCast [55], and AlphaPre [24], all under the shared evaluator so that every number is directly comparable.

We report the standard meteorological detection and fidelity metrics. For a binarization threshold τ , counting hits (TP), misses (FN), and false alarms (FP) over the thresholded prediction and ground truth, the Critical Success Index and Heidke Skill Score are

$$\text{CSI}_\tau = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}}, \quad \text{HSS} = \frac{2(\text{TP} \cdot \text{TN} - \text{FP} \cdot \text{FN})}{(\text{TP} + \text{FN})(\text{FN} + \text{TN}) + (\text{TP} + \text{FP})(\text{FP} + \text{TN})}, \quad (10)$$

where TN counts correct negatives. CSI-M is the mean CSI over the dataset's thresholds, and we additionally report CSI at the two highest (heavy-rain) thresholds, the most operationally important and hardest regime. We also report SSIM and MSE for pixel fidelity. The primary metrics are CSI-M and the high-threshold CSI: they measure forecast skill where blur-prone methods fail, whereas SSIM and MSE alone reward the conditional-mean smoothing we aim to avoid.

5.4 Main results

Across the four benchmarks and six metrics each, PG-FMM outperforms AlphaPre on 18 of the 24 metrics (Tab. 1). It wins every metric on the two large benchmarks: on SEVIR it raises CSI-M from 0.3259 to 0.3614 and roughly doubles the heaviest-threshold CSI ($0.0545 \rightarrow 0.0925$), and on MeteoNet it again leads

on all six metrics (CSI-M 0.4239 vs. 0.3824). The gains concentrate at the heavy-rain thresholds, exactly where deterministic and residual-anchor models blur. On the two smaller benchmarks the advantage is partial but follows the same pattern: on Shanghai PG-FMM wins four of six metrics, including the heavy-rain CSI-40, SSIM, and MSE, and matches AlphaPre on mean CSI; on CIKM it wins the two heavy-rain thresholds (CSI-35 and CSI-40) and is second on HSS, while AlphaPre keeps a narrow lead on mean CSI. PG-FMM also outperforms the strongest generative baseline, DiffCast, on SEVIR (CSI-M 0.3614 vs. 0.3050). Because of the compute budget these numbers are single-seed; we therefore emphasize the size of the margins and the consistent heavy-rain pattern rather than a significance test.

5.5 Ablation study

We ablate the two modules of PG-FMM on SEVIR at the unified operating point (16-member PMM, $N=4$; Tab. 2). Removing the motion prior (A1, the pure-generative head) lowers CSI-M, the heaviest-threshold CSI, and HSS and raises MSE, confirming that the prior contributes both detection skill and pixel fidelity. Evaluating the frozen prior alone (A2) is worse on every detection metric and on pixel error, which shows the generative head is necessary. Together the two ablations justify the motion-generative decoupling: neither module is sufficient on its own.

Inference-time sweeps on the same checkpoint (Tab. 3) show that ensemble size monotonically improves CSI and HSS, and that sampling skill saturates quickly in the number of steps: $N=4$ recovers essentially all of the CSI of $N=8$ at lower cost and is our default, while even $N=1$ produces a usable forecast. This is the practical payoff of the flow-map formulation: few-step sampling makes the multi-member ensemble that drives the high-threshold gains affordable.

5.6 Qualitative results

On representative heavy-rain cases, PG-FMM preserves the sharp, high-intensity cores that the AlphaPre baseline smooths away, consistent with the high-threshold CSI gains; the prior keeps the convection on its observed track while the generative head restores intensity. For verification metrics we report the probability-matched-mean ensemble, whereas qualitative panels show a single sample, which is visually sharper. A gallery of cases, including failure modes, is provided in the supplementary material.

6 Conclusion

We addressed radar precipitation nowcasting in the high-intensity regime, where deterministic and residual-anchor generative models blur away the convective cells that matter most. We proposed PG-FMM, which decouples the predictable

Table 1: Comparison on four radar precipitation benchmarks under the AlphaPre evaluation protocol (shared evaluator, thresholds, and test splits; our model uses a 16-member probability-matched-mean ensemble). The two highest (heavy-rain) thresholds are 181/219 (SEVIR), 24/32 (MeteoNet), and 35/40 (Shanghai and CIKM). The Type column follows the AlphaPre benchmark taxonomy [24]: ND denotes models without an explicit dynamics module and D denotes models with one. **Bold** = best per column, underline = second best. Our method wins all six metrics on SEVIR and MeteoNet, four of six on Shanghai, and two of six on CIKM (18/24 overall).

		SEVIR						MeteoNet					
Model	Type	CSI-M $\uparrow$	CSI-181 $\uparrow$	CSI-219 $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$	CSI-M $\uparrow$	CSI-24 $\uparrow$	CSI-32 $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$
ConvGRU	ND	0.2903	0.0879	0.0350	0.3619	0.6100	368.34	0.3401	0.2990	0.1431	0.4667	0.7833	12.85
MAU	ND	0.3076	0.1071	0.0516	0.3863	0.6505	355.48	0.3233	0.2839	0.0997	0.4452	0.7897	12.92
SimVP	ND	0.3108	0.1106	0.0517	0.3924	0.6508	383.56	0.3351	0.3002	0.1130	0.4573	0.7804	13.45
FourCastNet	ND	0.2686	0.0717	0.0339	0.3355	0.5976	410.27	0.3027	0.2533	0.1085	0.4216	0.6450	15.05
Earthformer	ND	0.2892	0.0844	0.0245	0.3665	0.6633	360.11	0.3205	0.2884	0.1237	0.4491	0.7772	14.43
PhyDNet	D	0.3017	0.1040	0.0278	0.3812	0.6532	357.63	0.3384	0.3194	0.1366	0.4673	0.7823	14.48
EarthFarseeer	D	0.3004	0.0992	0.0413	0.3829	0.6327	388.91	0.3404	0.3170	0.1372	0.4726	0.7542	14.10
NowcastNet	D	0.2791	0.0770	0.0351	0.3512	0.6839	412.94	0.3427	0.3206	0.1598	0.4751	0.7879	15.64
DiffCast	D	0.3050	0.1300	0.0582	0.3996	0.6482	559.59	0.3512	0.3340	0.1808	0.4846	0.7887	17.93
AlphaPre	D	<u>0.3259</u>	<u>0.1332</u>	0.0545	<u>0.4110</u>	<u>0.6884</u>	<u>345.18</u>	<u>0.3824</u>	<u>0.3633</u>	<u>0.2002</u>	<u>0.5164</u>	<u>0.7968</u>	<u>12.74</u>
Ours	D	0.3614	0.1859	0.0925	0.4593	0.7291	315.67	0.4239	0.4066	0.2293	0.5583	0.8426	9.25

		Shanghai						CIKM					
Model	Type	CSI-M $\uparrow$	CSI-35 $\uparrow$	CSI-40 $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$	CSI-M $\uparrow$	CSI-35 $\uparrow$	CSI-40 $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$
ConvGRU	ND	0.3612	0.3163	0.2062	0.4899	0.7796	33.56	0.3091	0.2009	0.1259	0.4006	0.6507	37.13
MAU	ND	0.3983	0.3621	0.2417	0.5346	0.7195	30.40	0.3039	0.2054	0.1241	0.3928	0.6325	40.74
SimVP	ND	0.3850	0.3549	0.2382	0.5194	0.7795	34.40	0.3052	0.2044	0.1321	0.3955	0.6538	38.06
FourCastNet	ND	0.3571	0.3108	0.2073	0.4868	0.5598	32.10	0.2980	0.1849	0.1015	0.3801	0.4359	36.14
Earthformer	ND	0.3503	0.3178	0.1872	0.4844	0.7298	35.57	0.3077	0.2039	0.1369	0.4001	0.6267	36.49
PhyDNet	D	0.3654	0.3236	0.2176	0.4957	0.7751	36.41	0.3038	0.2052	0.1287	0.3931	0.6541	39.56
EarthFarseeer	D	0.3926	0.3608	0.2343	0.5330	0.5405	32.68	0.3000	0.2046	0.1259	0.3911	0.6373	39.87
NowcastNet	D	0.3953	0.3608	0.2450	0.5334	0.7902	33.56	0.2991	0.1940	0.1188	0.3865	0.6713	40.96
DiffCast	D	0.4089	0.3740	0.2606	<u>0.5476</u>	0.7879	36.35	<u>0.3159</u>	0.2009	<u>0.1457</u>	0.4085	0.6499	42.78
AlphaPre	D	0.4178	0.3854	<u>0.2615</u>	0.5534	<u>0.7951</u>	<u>28.02</u>	0.3194	<u>0.2068</u>	0.1416	0.4137	<u>0.6568</u>	35.18
Ours	D	0.4181	<u>0.3819</u>	0.2636	0.4181	0.7962	26.05	0.3149	0.2218	0.1540	<u>0.4115</u>	0.6252	42.18

advective motion from the uncertain small-scale detail and conditions a generative flow-map head on a frozen physical prior rather than summing a residual onto a deterministic backbone, so that the prediction does not inherit conditional-mean blur. Concretely, a Lagrangian advection prior supplies an interpretable motion forecast, and a Flow-Map Matching head renders sharp detail conditioned on it, with few-step sampling and a probability-matched ensemble at inference.

Under a unified protocol over four radar benchmarks, PG-FMM surpassed the strongest published baseline on 18 of 24 metrics, winning every metric on SEVIR and MeteoNet and the heavy-rain thresholds on Shanghai and CIKM, with the largest margins exactly where mean-seeking methods fail. Ablations confirmed that the motion prior and the generative head are each necessary.

Several directions remain open. The advection prior is frozen after pre-training, and jointly adapting it with the generative head may further improve long-lead skill. Multi-seed evaluation, event-specific assessment of extreme rainfall, probabilistic skill metrics such as CRPS at scale, and an extension from single-channel radar to multi-channel or three-dimensional observations are natural next steps.

Table 2: Component ablation on SEVIR (16-member PMM, four sampling steps). The frozen motion prior (A0 vs A1) improves detection skill and pixel error, while the prior alone (A2) loses on every detection metric and on MSE. Together the two ablations justify the motion-generative decoupling: neither module is sufficient on its own.

	Configuration	CSI-M $\uparrow$	CSI-181 $\uparrow$	CSI-219 $\uparrow$	HSS $\uparrow$	MSE $\downarrow$
A0	PG-FMM (full)	0.3614	0.1859	0.0925	0.4593	315.67
A1	– motion prior (pure gen.)	0.3572	0.1754	0.0964	0.4522	317.18
A2	motion prior only (no gen. head)	0.3552	0.1737	0.0859	0.4493	334.95

Table 3: Inference-time trade-offs on SEVIR. Ensemble size monotonically improves CSI; NFE=4 captures $\sim 99.8\%$ of the CSI of NFE=8 at lower cost and is our default. sps = samples/sec.

Sweep	Setting	CSI-M $\uparrow$	CSI-181 $\uparrow$	HSS $\uparrow$	MSE $\downarrow$
Ensemble	1	0.3394	0.1667	0.4335	366.80
	4	0.3559	0.1816	0.4529	327.32
	8	0.3595	0.1843	0.4570	319.65
	16	0.3614	0.1859	0.4593	315.67
NFE	1	0.3463	0.1590	0.4361	304.50
	2	0.3523	0.1698	0.4453	307.27
	4	0.3614	0.1859	0.4593	315.67
	8	0.3620	0.1898	0.4616	324.98

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